

Kuranishi theory

Shing Tak Lam

November 17, 2025

In [CS14; Szé10], the authors study Kuranishi theory for cscK manifolds. As it does not seem to be written down in the literature, we present the analogous theory for Hermite–Einstein vector bundles. We refer to [Kob14; BS22] for related results.

1 Kuranishi theory

Let (X, ω) be a compact Kähler manifold, $\mathcal{E} = (E, \bar{\partial})$ a holomorphic vector bundle. Let h be a hermitian metric on E , such that $(\mathcal{E}, h)$ is Hermite–Einstein.

Let $G = \text{Aut}(\mathcal{E})$. This is reductive, with maximal compact subgroup $K = G \cap \text{U}(E, h)$. We have an elliptic complex

$$\mathcal{A}^0(\text{End } E) \xrightarrow{\bar{\partial}} \mathcal{A}^{0,1}(\text{End } E) \xrightarrow{\bar{\partial}} \mathcal{A}^{0,2}(\text{End } E).$$

parametrising deformations of $\mathcal{E}$.

Lemma 1.1. *There is a linear K -action on $H^{0,1}(X, \text{End } \mathcal{E})$.*

Proof. The linear action is by conjugation, i.e. for $f \in K$ and $\alpha \in H^{0,1}(X, \text{End } \mathcal{E})$, we have

$$f \cdot \alpha = f \circ \alpha \circ f^{-1}.$$

Note we are using the Dolbeault theorem, so that we consider α as a $\bar{\partial}$ -harmonic $(0, 1)$ -form with values in $\text{End}(E)$. As f is holomorphic, so is $f \cdot \alpha$. Thus, all we need to consider is

$$\bar{\partial}^*(f \cdot \alpha).$$

This then follows by the Nakano identity, as we know that f is covariantly constant. □

In fact, as all we needed was that f was covariantly constant, this extends to a G -action.

Theorem 1.2. *There exists a neighbourhood $\mathcal{U}$ of 0 in $H^{0,1}(X, \text{End } \mathcal{E})$, and a holomorphic embedding*

$$\Phi : \mathcal{U} \rightarrow \mathcal{A}^{0,1}(\text{End } \mathcal{E}),$$

such that

- (i) $\Phi(0) = 0$,
- (ii) Φ is K -equivariant,
- (iii) if $u_1, u_2 \in \mathcal{U}$ are in the same G -orbit, and $\Phi(u_1)$ is integrable, then $\Phi(u_2)$ is integrable, and $\Phi(u_1)$ and $\Phi(u_2)$ are in the same G -orbit.
- (iv) Any $\alpha \in \mathcal{A}^{0,1}(\text{End } E)$ such that $\bar{\partial} + \alpha$ is integrable, with α sufficiently close to 0, is in the $\text{GL}(E)$ -orbit of some $\Phi(u)$, with $u \in \mathcal{U}$ and $\Phi(u)$ integrable.

We will first follow [Kur65], where we construct the map Φ and prove (iv). For this, we do not need the Hermite–Einstein condition. Note that in [Kur65], Kuranishi constructs a slice of the Diff-orbits of complex structures on a smooth manifold. Whereas in [CS14; Szé10], the authors consider a symplectic manifold, and construct a slice of the Ham-orbits of compatible almost complex structures. The latter causes additional technical difficulties.

We note that there isn't an analogue of being compatible with a symplectic form in the vector bundle setting. As such, we are able to follow Kuranishi's original approach.

Recall that we can identify Dolbeault operators with the affine space $\mathcal{A}^{0,1}(\text{End } E)$, and the Dolbeault operator $\bar{\partial} + a$ is integrable if and only if $N(a) = \bar{\partial}a + a \wedge a = 0$. We would like to find a *finite dimensional* slice of the integrable locus $N(a) = 0$. One choice is given by

$$V = \{a \in \mathcal{A}^{0,1}(\text{End } E) \mid \bar{\partial}^* a = 0, N(a) = 0\}.$$

Let $\square = \bar{\partial}^* \bar{\partial} + \bar{\partial} \bar{\partial}^*$ denote the $\bar{\partial}$ -Laplacian, and let G denote the Green's operator, and H the harmonic projection. For any $a \in V$,

$$\square a + \bar{\partial}^*(a \wedge a) = 0.$$

Applying G , and using the fact that $a = Ha + G\square a$, we get that

$$Ha = a + G\bar{\partial}^*(a \wedge a).$$

Thus, V is a subset of

$$W = \{a \in \mathcal{A}^{0,1}(\text{End } E) \mid a + G\bar{\partial}^*(a \wedge a) \in H^{0,1}(X, \text{End } E)\}.$$

Now consider the map

$$\begin{aligned} F : L_k^2(\mathcal{A}^{0,1}(\text{End } E)) &\rightarrow L_k^2(\mathcal{A}^{0,1}(\text{End } E)) \\ a &\mapsto a + G\bar{\partial}^*(a \wedge a). \end{aligned}$$

The derivative of F at 0 is the identity map, and so there exists a small ball U of $0 \in H^{0,1}(X, \text{End } E)$, and a holomorphic map $\Phi : U \rightarrow L_k^2(\mathcal{A}^{0,1}(\text{End } E))$, such that $F \circ \Phi = \text{id}$. By elliptic PDE theory, in fact Φ defines a map $U \rightarrow \mathcal{A}^{0,1}(\text{End } E)$.

By construction, Φ defines a homeomorphism onto its image, which is a neighbourhood of 0 in W . In particular, this means that if a is integrable, $\bar{\partial}^* a = 0$ and a is sufficiently small, then a lies in the image of Φ . As gauge transformations preserve integrability, to prove (iv) it suffices to show that for any a sufficiently small, there exists a gauge transformation $g \in GL(E)$, such that

$$\bar{\partial}^*(g \cdot (\bar{\partial} + a) - \bar{\partial}) = 0.$$

Define

$$\begin{aligned} \Psi : H^0(\text{End } E)^\perp \times L_k^2(\mathcal{A}^{0,1}(\text{End } E)) &\rightarrow H^0(\text{End } E)^\perp \\ (\sigma, a) &\mapsto \bar{\partial}^*(\exp(\sigma) \cdot (\bar{\partial} + a) - \bar{\partial}). \end{aligned}$$

Here, the $^\perp$ denotes the L^2 -orthogonal complement in appropriate Sobolev completions. The derivative of Ψ at $(0, 0)$ is

$$d\Psi_{(0,0)}(\sigma, a) = -\bar{\partial}^* \bar{\partial} \sigma + \bar{\partial}^* a = -\square \sigma + \bar{\partial}^* a.$$

In particular, as $\square : H^0(\text{End } E)^\perp \rightarrow H^0(\text{End } E)^\perp$ is an isomorphism, this is surjective with bounded right inverse. Thus, by the implicit function theorem, there exists neighbourhoods $\tilde{U}$ of $0 \in L_k^2(\mathcal{A}^{0,1}(\text{End } E))$ and $\tilde{V}$ of $0 \in H^0(\text{End } E)^\perp$, such that for all $a \in \tilde{U}$, there exists a unique $\sigma = \sigma(a) \in \tilde{V}$ such that

$$\Psi(\sigma(a), a) = 0.$$

That is, if we set $f = \exp(\sigma)$, and $f \cdot a = f a f^{-1} + f \bar{\partial} f^{-1}$, then $\bar{\partial}^*(f \cdot a) = 0$. This proves (iv).

For (ii), we note that F is K -equivariant, as the Green's operator and $\bar{\partial}^*$ commutes with the K -action. Thus, so is Φ .

For $u \in U$, let $a = \Phi(u)$, so $F(a) = u$. In particular, this means that $u = F(a) = a + G\bar{\partial}^*(a \wedge a)$. Now

$$\bar{\partial}^* a = -\bar{\partial}^* G \bar{\partial}^*(a \wedge a) = -(\bar{\partial}^*)^2 G(a \wedge a) = 0,$$

where we used the fact that $\bar{\partial}^* u = 0$ and that $\bar{\partial}^* G = G \bar{\partial}^*$. Moreover,

$$\begin{aligned} N(a) &= \bar{\partial} a + a \wedge a \\ &= -\bar{\partial} G \bar{\partial}^*(a \wedge a) + a \wedge a \\ &= -G(\bar{\partial} \bar{\partial}^*)(a \wedge a) + H(a \wedge a) + G(\bar{\partial}^* \bar{\partial})(a \wedge a) + G(\bar{\partial} \bar{\partial}^*)(a \wedge a) \\ &= H(a \wedge a) + G(\bar{\partial}^* \bar{\partial})(a \wedge a) \\ &= H(a \wedge a) + 2G \bar{\partial}^*(N(a) \wedge a). \end{aligned}$$

Since the images of G and H are orthogonal, this means that $N(\alpha) = 0$ if and only if $H(\alpha \wedge \alpha) = 0$. Thus, a neighbourhood of 0 in V is an analytic set contained in the image of Φ .

The result of (iii) follows by local G -equivariance, and the G -invariance of integrability.

References

- [BS22] Nicholas Buchdahl and Georg Schumacher. “Polystable Bundles and Representations of Their Automorphisms”. In: *Complex Manifolds* 9.1 (2022), pp. 78–113. issn: 2300-7443. doi: 10.1515/coma-2021-0131.
- [CS14] Xiuxiong Chen and Song Sun. “Calabi Flow, Geodesic Rays, and Uniqueness of Constant Scalar Curvature Kähler Metrics”. In: *Annals of Mathematics. Second Series* 180.2 (2014), pp. 407–454. issn: 0003-486X,1939-8980. doi: 10.4007/annals.2014.180.2.1.
- [Kob14] Shoshichi Kobayashi. *Differential Geometry of Complex Vector Bundles*. Princeton Legacy Library. Princeton University Press, Princeton, NJ, 2014. isbn: 978-0-691-60329-2.
- [Kur65] M. Kuranishi. “New Proof for the Existence of Locally Complete Families of Complex Structures”. In: *Proc. Conf. Complex Analysis (Minneapolis, 1964)*. Springer, Berlin-Heidelberg-New York, 1965, pp. 142–154.
- [Szé10] Gábor Székelyhidi. “The Kähler-Ricci Flow and K-polystability”. In: *American Journal of Mathematics* 132.4 (2010), pp. 1077–1090. issn: 0002-9327,1080-6377. doi: 10.1353/ajm.0.0128.